Core Pure 2 · Chapter 5

Polar Coordinates

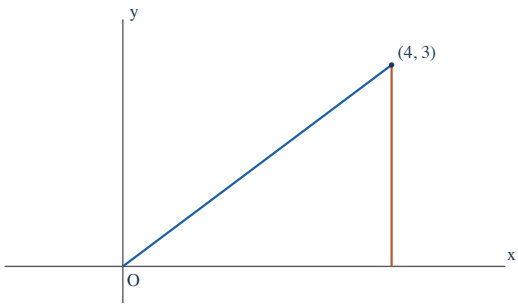
Conversions · Curves · Areas · Tangents

What the devil are polar coordinates?

Luisa · 2

A polar point (r, θ) is a distance r from the pole, at angle θ anticlockwise from the initial line. Angles are in radians.

Convert the Cartesian point $(4, 3)$ to polar form.



Answer

Converting to/from polar coordinates — Luisa · 3

Converting coordinates

$x = r\cos\theta,\ y = r\sin\theta,\ r^2 = x^2 + y^2$. Choose the angle from the correct quadrant.

Convert these Cartesian coordinates to polar form:
$(0, 2),\ (-3, 0),\ (1, 1),\ (-5, 12),\ (3\sqrt{3}, -3)$.

Polar coordinates

Polar to Cartesian equations — Luisa · 4

Convert to Cartesian form:

a. $r = 5$

b. $r = 2 + \cos 2\theta$

c. $r^2 = \sin 2\theta$

a

b

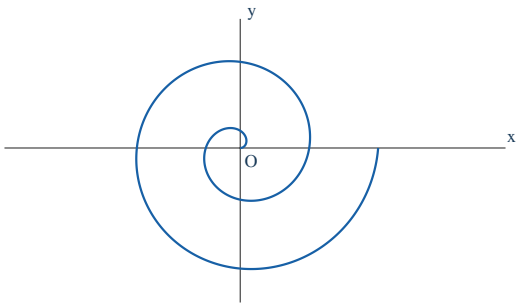
c

Why polar form sometimes are preferable — Luisa · 5

A spiral: $r = \theta, \theta \geq 0$.

As the angle increases, the distance from the pole increases.

Its parametric form is $x = \theta \cos\theta,\ y = \theta \sin\theta$.



Why keep polar form?

Cartesian to polar equations — Luisa · 6

Convert to polar form:

a. $y^2 = 4x$

b. $x^2 - y^2 = 5$

c. $\sqrt{3}y = x + 4$

a

b

c

2. Convert these polar coordinates to Cartesian coordinates:

a. $(6, \pi/6)$

b. $(6, -\pi/6)$

c. $(6, 3\pi/4)$

a

b

c

3a. Convert $r = 2$

3b. Convert $r = 3\sec\theta$

3c. Convert $r = 5\csc\theta$

a

b

c

3d. Convert $r = 4a\tan\theta\sec\theta$

3e. Convert $r = 2a\cos\theta$

3f. Convert $r = 3a\sin\theta$

d

e

f

4a. Convert $x^2 + y^2 = 16$

a

4b. Convert $xy = 4$

b

4c. Convert $(x^2 + y^2)^2 = 2xy$

c

4d. Convert $x^2 + y^2 - 2x = 0$

d

4e. Convert $(x + y)^2 = 4$

e

4f. Convert $x - y = 3$

f

5. Prove the distance formula for polar points $P = (r_1, \theta_1)$, $Q = (r_2, \theta_2)$.

Answer

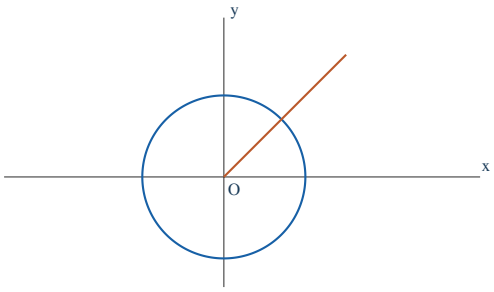
Recognising polar curves

$r = a$: circle, centre the pole, radius a.

$\theta = \alpha$, $r \geq 0$: a ray from the pole.

$r = a\theta$, $\theta \geq 0$: an Archimedean spiral.

Plot points and mark the direction of increasing angle.



Sketch $r = \sin 4\theta$, using $r \geq 0$.

Start with $0 \leq \theta \leq \pi/4$.

Plot the loop

Sketch

Sketching using tables of values — Luisa · 13

Sketch the cardioid $r = a(1 + \cos\theta)$, $a > 0$.

Key points

Sketch

More sketchies — Luisa · 14

Sketch:

a. $r = a\sin 3\theta$, $r \geq 0$.

b. $r^2 = a^2 \cos 2\theta$.

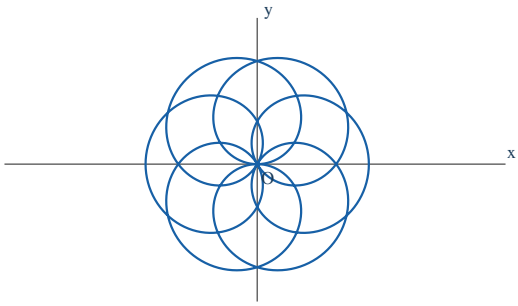
Answer

Sketch

Exploration

Plot $r = \cos(3\theta/4)$, allowing signed radii, for $0 \le \theta \le 8\pi$.

A negative radius is plotted in the opposite direction. This extension is separate from our usual $r \ge 0$ convention.

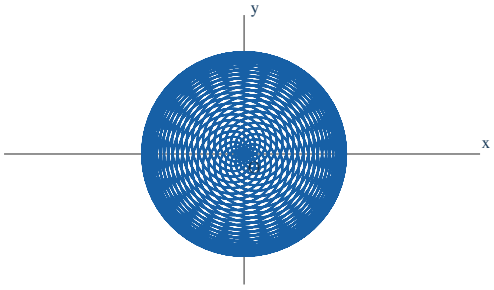


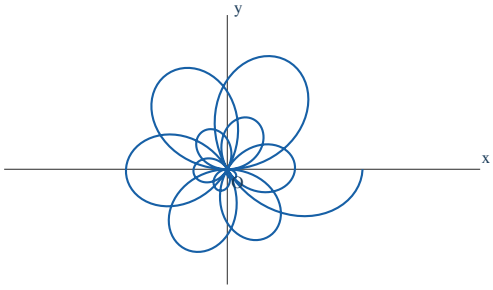
Observation

More polar patterns

Compare $r = \cos(13\theta/99)$ with $r = \theta\cos(3\theta/2)$, allowing signed radii.

The first produces a closed multi-loop pattern; the second expands outward as the angle increases.

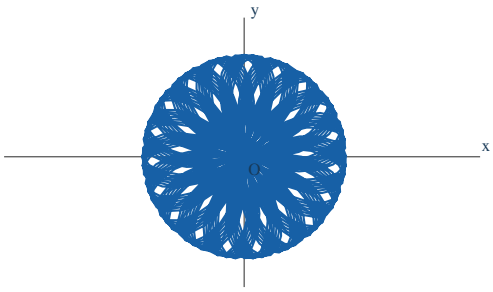




An irrational frequency

Plot $r = \cos(e\theta)$ for $0 \leq \theta \leq 50\pi$, allowing signed radii.

Since e is irrational, there is no common angular period that returns the full parametrisation to its starting state. The pattern does not close exactly.



Limaçons

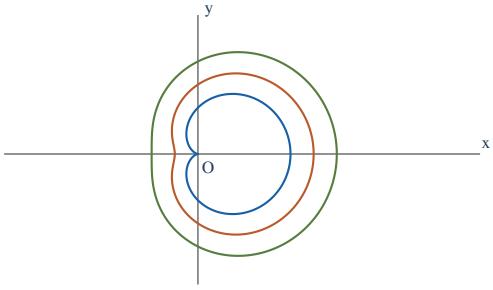
$r = a(p + q\cos\theta)$, with $a, q > 0, p \geq q$.

$p = q$: cardioid with a cusp.

$q < p < 2q$: dimpled limaçon.

$p \geq 2q$: convex limaçon.

The curve is symmetric in the initial line.



Egg vs Dimple — Luisa · 19

Sketch and compare:

a. $r = a(5 + 2\cos\theta)$

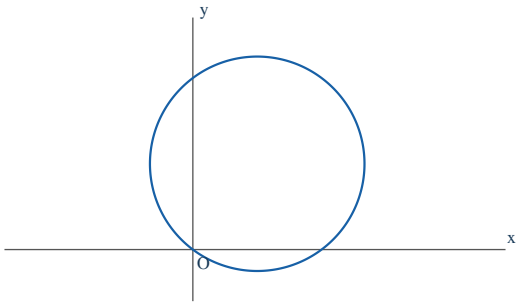
b. $r = a(3 + 2\cos\theta)$

Answer

Sketch

Final Example — Luisa · 20

The locus $|z - 3 - 4i| = 5$ is a circle. Find its polar equation.



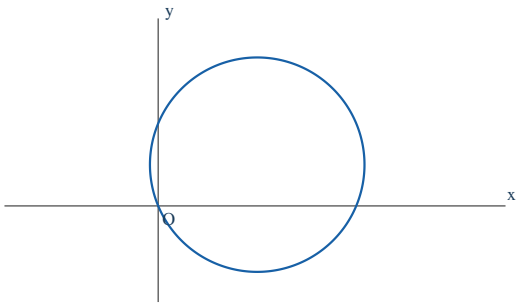
Answer

2. Sketch $r = k\sec(\pi/4 - \theta)$, $k > 0$, giving the axis intercepts.

Answer

3a. Sketch $|z - 12 - 5i| = 13$ on an Argand diagram.

b. Show its polar equation is $r = 24\cos\theta + 10\sin\theta$.



a

b

Curve summary

$r = a$: circle at the pole.

$\theta = \alpha$: ray at angle α.

$r = 2a\cos\theta$: circle centred at $(a, 0)$, radius a.

$r = k\theta$: spiral.

Always state the angular interval and whether signed radii are permitted.

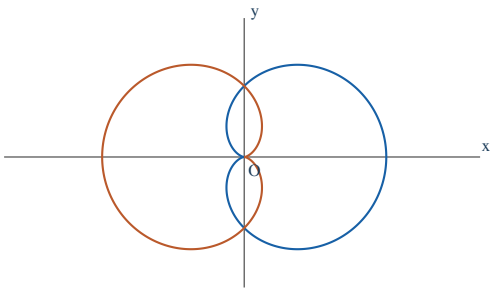
Symmetry and orientation

$a(1 + \cos\theta)$: cardioid facing right.

$a(1 - \cos\theta)$: cardioid facing left.

Replacing cosine by sine rotates the orientation.

$a(3 + 2\cos\theta)$: a dimpled limaçon, with minimum radius a.



More standard forms

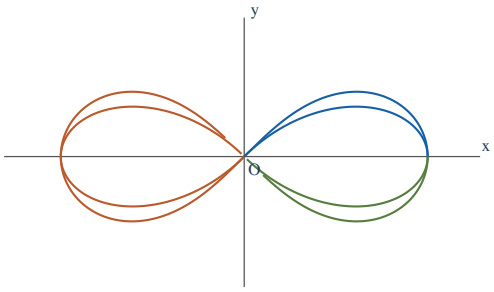
$r = a\cos 2\theta$, $r \geq 0$: two positive-radius petals.

$r^2 = a^2 \cos 2\theta$: lemniscate.

$r = p\sec(\alpha - \theta)$: straight line

$x\cos\alpha + y\sin\alpha = p$.

In particular, $r = a\sec\theta$ gives $x = a$.



From strips to sectors

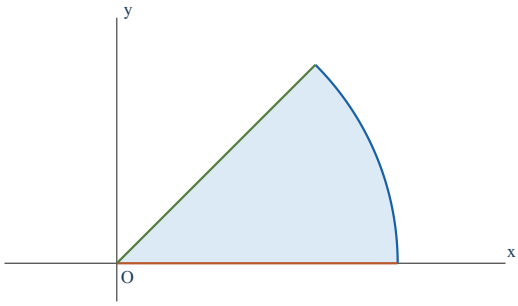
In Cartesian coordinates, area is accumulated from thin strips: $dA = y\,dx$.

A solid of revolution uses thin discs: $dV = \pi y^2 dx$.

For polar area, use thin sectors with angle $d\theta$.

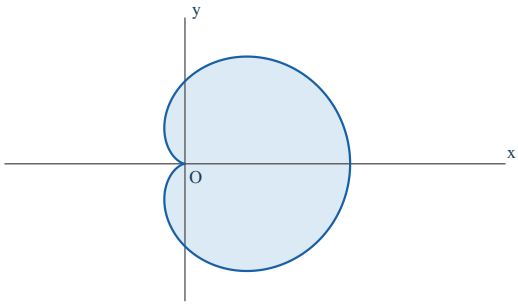
Area in polar coordinates

A sector of radius r and angle $d\theta$ has area approximately $\frac{1}{2}r^2 d\theta$.



Formula

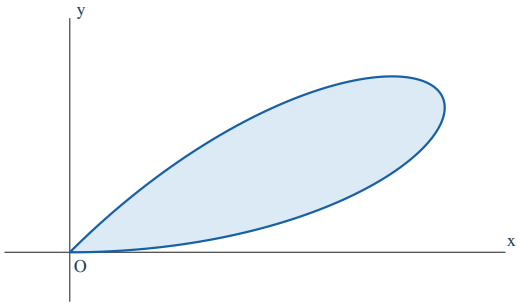
Find the area enclosed by $r = a(1 + \cos\theta)$.



Answer

Another Example

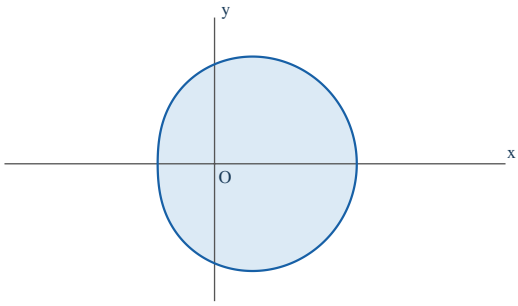
Find the area of one loop of $r = a \sin 4\theta$, $r \geq 0$.



Answer

Test Your Understanding So Far

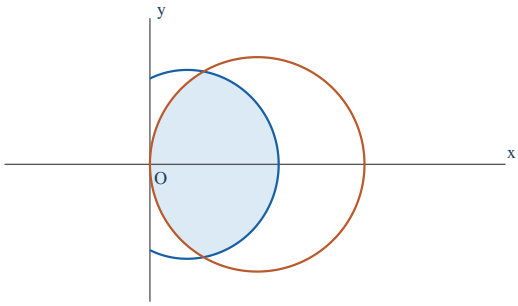
The area enclosed by $r = a + 3\cos\theta$, $a > 3$, is $107\pi/2$. Find a.



Answer

Intersecting Areas — Luisa · 31

Find the area common to $r = 2 + \cos\theta$ and $r = 5\cos\theta$.

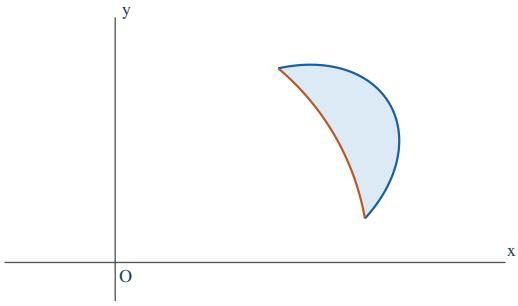


Intersections

Area

Test Your Understanding — Luisa · 32

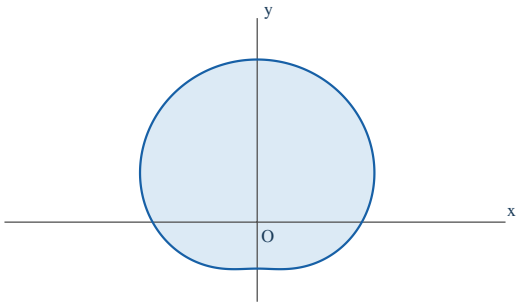
Find the area outside $r = 2$ but inside $r = \frac{3}{2} + \sin 3\theta$, for $0 \le \theta \le \pi/2$.



Intersections

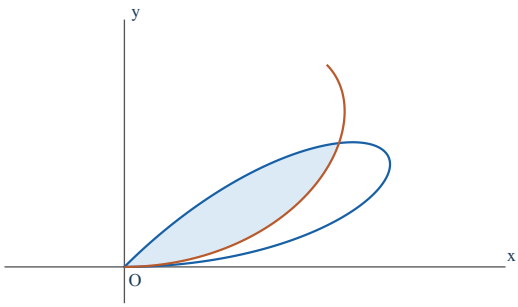
Area

4. The area enclosed by $r = a + 5\sin\theta$, $a > 5$, is $187\pi/2$. Find a.



Answer

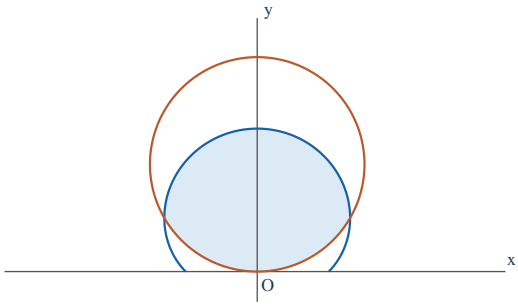
5. Find the finite area common to $r = a\sin 4\theta$ and $r = a\sin 2\theta$, with $0 \leq \theta \leq \pi/2$ and $r \geq 0$.



Intersections

Area

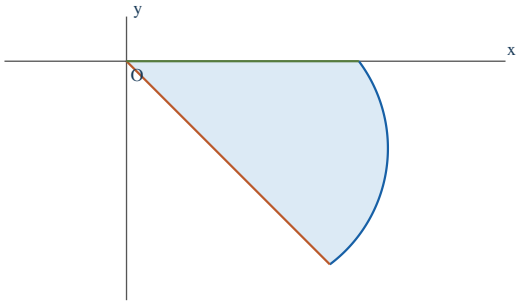
6. Find the area common to $r = 1 + \sin\theta$ and $r = 3\sin\theta$.



Intersections

Use symmetry

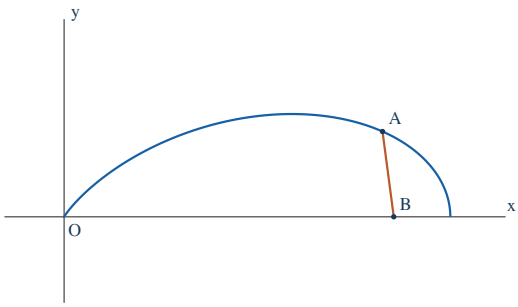
7. Sketch the region $-\pi/4 \leq \arg z \leq 0$, $|z - 4 + 3i| \leq 5$. Find its area to 3 significant figures. The circle has polar equation $r = 8\cos\theta - 6\sin\theta$.



Sketch

Area

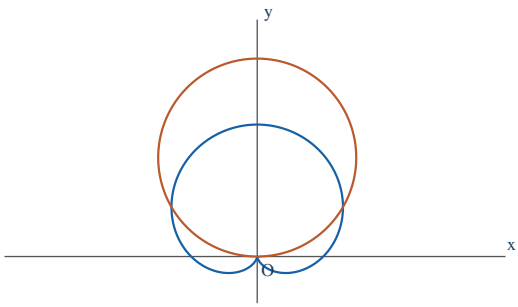
9. On $r = 1 + \cos 3\theta$, $0 \leq \theta \leq \pi/3$, point A has radius $(2 + \sqrt{2})/2$. Point B is on the initial line with the same radius. Find the area between arc A to $(2, 0)$, AB and the initial line, to 3 significant figures.



Sector

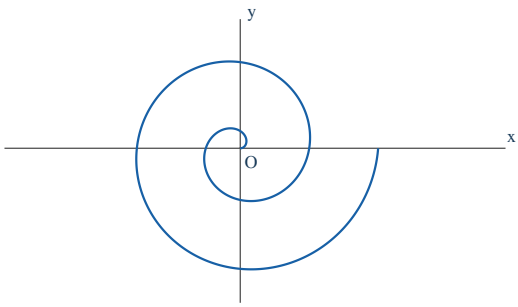
Subtract triangle

10. Find the area inside the cardioid $r = 1 + \sin\theta$ and outside the circle $r = 3\sin\theta$, to 2 decimal places.



Answer

Challenge. A shell is modelled by $r = k\theta$, $0 \leq \theta \leq 4\pi$. The width along the initial line, from the point at 3π to the point at 4π, is 3 cm. Find k and the area inside the outer turn.



Scale

Area

Tangents to polar curves

$x = r\cos\theta, \qquad y = r\sin\theta$

Differentiate

Example — Luisa · 41

Find the regular points where $r = a(1 + \cos\theta)$ has horizontal tangents.

Answer

Sketch

Test Your Understanding So Far — Luisa · 42

For $r = 1 + 2\cos\theta$, $0 \leq \theta \leq \pi/2$, find the radius at the point with a horizontal tangent.

Answer

Sketch

Example

Find horizontal tangents to $r = a \sin 2\theta$, $0 \leq \theta \leq \pi/2$.

Answer

Sketch

Example

Find vertical tangents to $r = a \sin 2\theta$, $0 \leq \theta \leq \pi/2$.

Answer

Sketch

Proof of dimple vs egg — Luisa · 45

Explain why $r = a(p + q\cos\theta)$, $p > q > 0$, has a dimple when $p < 2q$.

Answer

Exercise 5D - Core Pure Year 2 Pages 115-116 — Luisa · 46

4. Find the polar coordinates on $r = a(7 + 2\cos\theta)$ where the tangents are parallel to the initial line.

Answer

Sketch

5. Find the tangent lines to $r = 2 + \cos\theta$ that are perpendicular to the initial line.

Answer

Sketch

6. Find the point on $r = a(1 + \tan\theta)$, $0 \leq \theta < \pi/2$, with a tangent perpendicular to the initial line.

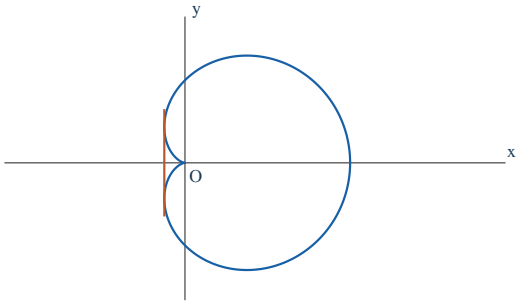
Answer

7. A point A on $r = 1 + 3\cos\theta$, $0 \leq \theta \leq \pi/2$, has a horizontal tangent. Find the exact length OA.

Answer

Sketch

8. For the cardioid $r = 2(1 + \cos\theta)$, find the area between the curve and its left-hand vertical common tangent, to 3 significant figures.



Tangencies

Area